Bookerfly: a Community-Driven and Collaborative Reading System

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ABSTRACT
BookerFly is a reading platform which focuses on making the reading experience fun and engaging. Readers can easily interact and engage with one another through group readings, shared annotation or reading activity statistics without having to remove themselves from the reading space and breaking the immersion. In this pictorial, we present the design process of BookerFly, and we discuss how to further expand our design in order to build a more collaborative reading experience.

Author Keywords
Collaborative reading



CSS Concepts
• Human-centered computing ~ *Interaction design*

INTRODUCTION
Popular digital reading platforms like Wattpad, Webtoon, Apple Books or even Archive of Our Own (AO3) have a wide user base and can be considered "leaders" in the reading community. However, despite the many features they provide, these systems still have limitations in terms of community engagement and collaboration. Readers often wish to discuss specific passages, react to moments in the story, or share quotes with others. In many cases, this requires leaving the reading environment and switching to external messaging platforms, which disrupts the reading experience.

In order to identify the actual existing limitations of current reading platforms, we conducted eight story interviews with frequent readers: six university students and two young adults. All participants reported reading on a daily basis and most of them used multiple devices (phone, tablet, and laptop) to access digital books. Participants also reported reading various formats, including novels, comics, and web-based stories.

From these interviews, we identified the following issues, which we wanted to focus on:

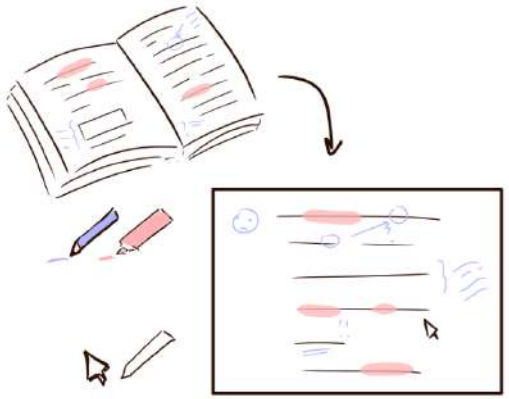
- Readers enjoy engaging with others, through sharing their opinions, reacting to specific scenes, giving reviews and discussing with each other. However, in current existing systems, users are usually forced to leave the reading space if they want to further engage with the community, which breaks the immersion.
- Readers lack ways to interact and engage with the story directly within the context. As of now, such ways generally only include bookmarks or highlights.

To answer these problems, we introduce features that aim to enhance the reading experience. Our system is mainly aimed at students and young people, who often read frequently and are comfortable using digital technologies. It is designed for PC and tablets, so that users can access BookerFly anytime and anywhere.

Solution overview

BookerFly is built around the following features, focused on making the overall experience more social and engaging for users:

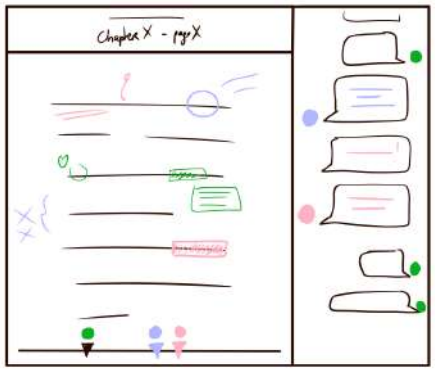
- *Annotations* that allow the readers to highlight, draw or write notes directly on the page as they would do on a real paper book. By simply selecting a text, they can directly highlight it and add text or drawings to it. Users can also add freehand annotations wherever on the page.



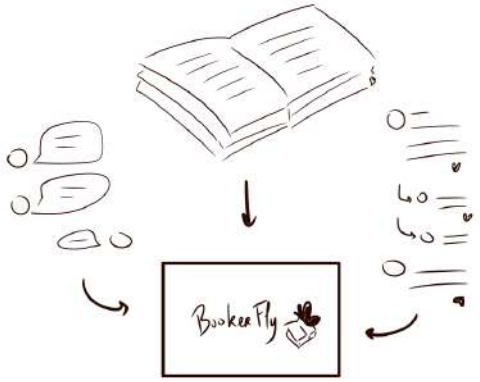
- *Snippet sharing*, which enables users to save and share meaningful excerpts. To do this, they just need to select the relevant text, then click on the Snippet option in the menu. Then, they can save it to a Snippet collection, which they can revisit later, or directly share it to a chat. Snippets act as hyperlink: clicking on them brings the user directly to the excerpt within its original context. If the receiver's reading progress is not yet at the point where the Snippet appears in the story, it shows up as a spoiler, which they can choose to view or not.



- *Group reading sessions*, where multiple readers can read, annotate and discuss a book together. Users can asynchronously read the same book and share their reading experience with one another. Each user's annotations are visible to the others, and a shortcut to open the relevant group chat is accessible on-screen.



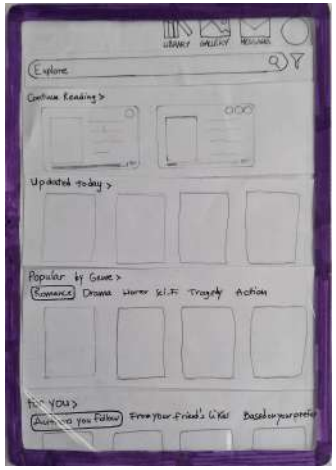
- *Integrated messaging and comments*, supporting interaction between readers without leaving the reading interface.



- *Reading statistics*, which provide the user meaningful insights into reading habits over time. Various statistics are available, including but not limited to reading speed, daily-weekly-monthly reading activity, and preferred genres. Group statistics are also available in Group Reads, in order to make the overall reading experience more engaging.



SCENARIO



Anne is a university student who loves reading during her daily commute. She opens BookerFly on her tablet to continue reading her book.

Fig 1: *The Home page of Bookerfly*



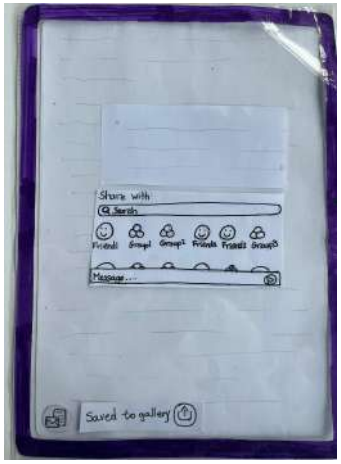
She picks up a Group Read with her two friends from the "Continue Reading" section on the homepage.

Fig 2: *Access to a Group read book*



She sees an excerpt that surprises her and selects it, and clicks on the last menu option "Snippet".

Fig 3: *Selecting text*



Anne shares the text Snippet to a group chat.

Fig 4: *Share the selected text in the group chat*



She sees the Snippet that she just sent appear in the group chat.

Fig 5: *Result in chat box and view of other opinion sharing*



Anne decides to add annotations, which will be visible to her other friend in the same Group Read.

Fig 6: *Add annotations and drawings*

DESIGN PROCESS

Through brainstorming, constrained ideation, and iterative prototyping, we explored different possibilities before defining the core features of our system.

Early brainstorming

Before starting the design process, we first conducted a brainstorming session to explore different ways to make digital reading platforms more collaborative and interactive, as most current reading applications focus on individual reading experiences and limited opportunities for interaction between readers.

To structure our ideation process, we used the 10+10 technique, which helped us come up with our core features. We came up with several ideas but decided to focus on two main dimensions :

- Interaction with the content itself, such as highlighting, annotating, or reacting to passages and images.
- Interaction with other readers, through discussions, shared reading experiences, and collaborative engagement.

Design

After identifying the most relevant ideas, we moved to the design phase, where we refined and structured the main features of our platform.

Annotations and Snippet sharing

The first feature we developed focuses on enabling readers to directly interact with the content displayed on the screen. Reading is often a reflective activity, and many readers enjoy highlighting passages, taking notes, or saving memorable moments from a story. Our system therefore integrates tools that allow users to actively engage with the text or images.

Users can :

- Write, sketch or draw private annotations directly on the page they are reading
- Take, save and share snippets of the story (excerpts that can be text or image) to direct messages or group chats within the app.

The snippet functionality allows readers to easily share moments they found meaningful, surprising, or emotionally impactful. Moreover, to prevent spoilers from negatively affecting the reading experience, the system includes a spoiler detection mechanism. If a reader receives a snippet from a section of the story they have not yet reached, the system automatically flags it as a spoiler and hides its content until the user chooses to reveal it.

Reading statistics

The second feature focuses on helping users reflect on their reading habits over time. We introduce reading statistics summaries which can be on

- a weekly
- monthly
- or yearly basis

These statistics may include metrics such as the amount of time spent reading, the number of pages or chapters completed, and overall reading consistency. The system can also highlight the genres that the reader engages with most frequently. Users can choose whether to display these summaries publicly on their profile or keep them private for personal tracking.

Prototyping the interface

The next stage of the design process involved developing paper prototypes of the interface and interaction flow.

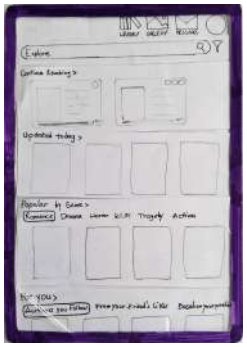
We began sketching initial ideas of the overall layout and visual structure of the interface. We decided to take inspiration from existing platforms when designing the main page, to ensure usability and familiarity. This decision helped us maintain recognizable navigation patterns while integrating our collaborative features.



Apple books interface



Wattpad interface



Bookerfly’s home page

For the group reading feature, we introduced the concept of a group chat bubble embedded within the group book interface. This design choice allows readers participating in the same group reading session to easily discuss passages, react to events in the story, or share snippets without leaving the reading environment.



The chat can be accessed by clicking an arrow located on the side of the screen, allowing users to open or collapse the conversation panel as needed. This ensures that the discussion remains easily accessible while minimizing visual obstruction of the reading view.

To add annotations, the user activates annotation mode by tapping twice on the screen. This interaction clears the page of interface elements and reveals a pen menu that consists of a pen, a highlighter and an eraser, allowing the user to draw or write directly on the page.



DISCUSSION

Suggested evaluation :

We ran a user study with paper prototypes. We had 3 participants who all had to do the same task: once with our first prototype (condition 1) and once with our second prototype (condition 2).

Each participant was asked to :

- Open an ongoing Group Read
- Annotate a page using the following ways :
  - Highlight
  - Write something
  - Draw
- Take a new Snippet and then share it

The order of the condition was changed between each participant to counter the order effect.

We gathered interesting insights from this study. However, the fact that we used a paper prototype as well as the Wizard of Oz technique to simulate features and system responses may have disturbed the study. We would like to build a higher fidelity prototype and run a similar experiment, with a few changes to the protocol and more importantly a higher number of participants

Future work :

From this user study, we found issues in our system and propose the following improvements :

Changing annotation interactions and making them more intuitive.

Add a Help button (in the toolbar): when clicking on it, a short tutorial appears to explain which actions the user can do on the text.

Selecting text should also open annotation mode, this time for annotating specific parts of the text (just like we did for our 2nd Prototype). Clicking on the Pen icon (in the menu above the selection) opens a Drawing Annotation Panel, whereas clicking on the Text icon opens a Text Annotation Panel and the keyboard. These Annotations appear as little markers next to the associated selections, and the panels can be shown or hidden by clicking.

Changing Snippets to make them more discoverable and understandable

Change the Snippet icon (when user selects a text) to a Scissors Icon for more clarity.

When clicking on the Snippet tool, add an explanation at the top of the Snippet : “This is a Snippet of your chapter: save it to your Snippet Collection or share it to your friends!”.

Remake the Snippets Gallery into a Snippets Collection: add a description (or something similar) at the top of this screen: “Snippets Collection : Here, you can store and view your favorite extracts, also known as Snippets, from your book collection. Click on a Snippet to view it, delete it, share it to your friends or go to the specific chapter it originates from!”.

Remove Image Snippets since we’re not currently tackling mangas, comics or graphic novels.

Making all the icons/buttons bigger and more explicit

Participants found it hard to understand the exact role of all buttons. Some were too similar to each other, and some had icons too vague. As our system uses menu icons for important features, it needs to have explicit buttons for users to easily find whenever it is needed.

WORK DISTRIBUTION

Every member contributed equally to this project.